

FOUR INSTRUMENTS. PHYSICAL EXPRESSION.

Maremba

Player guide / 1.4.0f6

Maremba is a physically modelled percussion instrument by Protocolus. Its four instruments generate each note from vibrating bars or tines, their supports and resonators. There are no sampled instrument notes. This guide covers **1.4.0f6**.



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Start playing

1. Create Maremba and use its main stereo outputs. The default patch is **Concert Grand Rosewood**.
2. Play a short phrase at several velocities. Softer and harder strokes change the physical contact and tone as well as the level.
3. Choose an instrument, then a Striker. Adjust **Hardness** and **Position** while playing new notes to shape the attack.
4. Use **Close**, **Room** and **Piezo** to balance the sound. **Volume** is the overall level; the small **VOL** knobs beneath playing controls are velocity sensitivities.
5. Start with **16 Voices** and **2x Quality**, the factory-bank settings. Use the native preset controls to explore and save sounds; the device-name label is editable.

The 20 factory presets all start with **Buzz and its VOL at zero**. The other nine small VOL controls start at **10%**. A preset's name can describe its inspiration: **Birifor Gylil Buzz** also starts clean, and its Buzz knob adds the membrane sound.

Choose an instrument

Instrument	Character	Striker choices
Imperial Rosewood 5.0	Concert-marimba bars and tuned tube resonators; a rounded attack and sustained bloom	Yarn, Cord, Rubber, Wood, Shaft
Mayan Padauk 4.3	Shorter wooden bars and box resonators; Buzz adds charleo membranes	Yarn, Cord, Rubber, Wood, Shaft
Balafon Ancestral	Wooden keys and gourd resonators; Buzz adds mirliton membranes	Yarn, Cord, Rubber, Wood, Shaft
Kalimba Artisan	Plucked steel tines, a yielding bridge and shared wooden box; Buzz adds board rattles	Flesh, Thumb, Nail, Pick, Tap

These are voiced instrument families. All accept the full MIDI note range; playing far outside an acoustic instrument's register intentionally explores the model beyond that instrument's usual range.

Shape the strike

Control	What it changes
Striker	The contacting material or thumb technique. Shaft strikes with the mallet handle; Kalimba Tap uses a fingernail impact.
Hardness	Mallet stiffness, or the Kalimba thumb's release speed. Higher values generally produce a firmer attack.
Position	The contact point on a bar, or the thumb position near the tine tip. Passing a vibration node can remove particular overtones, so the change is not simply brighter or darker.
Decay	Nominal ring-time control from 0.1 to 8; 1 is the instrument's natural decay, rather than one second. Kalimba keeps short settings unchanged, then gradually increases from natural sustain at 1 to twice natural sustain at 8.
Variance	The spread of contact stiffness, speed, landing position and damping from stroke to stroke. Zero retains subtle natural variation; higher values make the differences more pronounced.
Artifacts	Subtle rebounds, brushes and support collisions. Their timing and shape vary continuously. Set Artifacts and its VOL to zero to disable these events; natural strike variation remains.

Repeated notes interact with the existing vibration. A re-struck bar is already moving when the mallet meets it; a Kalimba re-pluck first damps the tine. Equal note and velocity values therefore do not force identical strokes. Explicit audio Reset restarts the deterministic variation sequence for reproducible renders.

Resonance and character

Control	What it changes
Carving	At 100%, the voiced instrument geometry. Turning down approaches a plain bar on the wooden instruments and a stiffer tine/support profile on Kalimba. The fundamental remains tuned to the note.
Coupling	Interaction with the tube, box or gourd air. On Kalimba it also controls bridge loading and how strongly the tine drives the shared box.
Bloom	Frame/support resonance on the wooden instruments and shared box/soundboard radiation on Kalimba.
Sympathetic	Other bars or tines ringing through their shared support or bridge.
Buzz	Tube/box/gourd membranes on the wooden instruments; loose board buzzers on Kalimba. Set both Buzz and its VOL to zero to remove this layer.
Tension	Support damping even with Buzz at zero. Higher tension damps upper modes more strongly. With Buzz raised it also changes membrane threshold and tone, or Kalimba buzzer preload. The centre is the instrument's nominal tension.

Membranes and buzzers need enough excitation to sound. Firm nail/pick plucks bring out Kalimba board rattles; the Balafon's top keys from MIDI 84 have no membranes. These are physical response differences rather than inactive controls.

Velocity response

Velocity always enters the physical excitation. On wooden bars it changes mallet speed and the resulting force/contact history. On Kalimba it changes the thumb's held force and tine deflection; softer strokes also release the tine more slowly. This response remains when every VOL sensitivity is zero.

The four **Velocity Response** switches select the following curves. **Sigmoid** is the device default and is selected in all 20 factory presets.

Response	Playing feel
Soft	Light strokes reach stronger dynamics sooner.
Linear	The direct reference response.
Hard	Strong dynamics require a firmer touch.
Sigmoid	An S-shaped curve with gentler response near its ends and stronger contrast through its middle.

Sigmoid is the same smoothstep curve previously named **Expressive**, at the same stored index 3. Older user patches keep their saved curve choice.

Each small **VOL** knob adds velocity sensitivity to its associated control: Hardness, Position, Decay, Carving, Coupling, Bloom, Sympathetic, Buzz, Tension or Artifacts. These are not audio volume controls. The main knob sets the starting value, and VOL adds a fraction of that control's full range:

VOL	MIDI velocity 1	MIDI velocity 127
0%	No change	No change
50%	No change	Adds half the knob range
100%	No change	Adds the full knob range

The result stops at the main knob's maximum. The selected Velocity Response shapes the progression between these endpoints. For example, Hardness at 20% with VOL at 50% reaches 70% at velocity 127. Decay spans 0.1-8 nominal control units, so half its range adds 3.95 before the instrument's sustain response. Kalimba applies its shorter range after this addition, so even full Decay VOL cannot exceed twice natural sustain. Buzz VOL can add buzz even when the main Buzz knob is zero; the factory presets keep both at zero.

Bloom and Sympathetic apply each strike's sensitivity to its drive into the shared resonances. Buzz and Tension retain each strike's sensitivity while their main knobs move. The other strike controls take effect on new strokes, including repeated notes. Presets store Buzz VOL at zero and the other nine depths at 10%. Loading old patches without these fields uses the device defaults of 20% for Hardness and Decay, and zero for the other eight. Older patches with nonzero VOL settings intentionally sound different under this additive calibration; their stored knob values are preserved.

Playing techniques

On the wooden instruments, raise the **mod wheel** to roll held notes with alternating mallets. More wheel increases the roll rate. Release the notes or lower the wheel to end the roll; the sound can ring on. On Kalimba, the wheel covers the box's sound hole and changes its resonance. **Pitch bend** spans ± 2 semitones and also bends ringing notes.

Rear **Note Off** offers **Ring**, **Damp** and **Dead**. Ring leaves the instrument sounding after key release. Damp brings a hand or thumb onto the released key. Dead keeps the mallet against the bar from the stroke, or brings the Kalimba thumb back shortly after its pluck. Stopping Reason's transport damps sounding keys and sympathetic ringing.

Capture, room and output

Close blends the close microphones. **Room** blends the far microphones and room return. **Piezo** blends a contact pickup attached to the instrument; it responds to local bending, bridge motion and mounting impacts, then passes through the sensor's electrical load. On Kalimba, the pickup also senses the box plate at its mounting point. This gives it a different attack and overtone balance from Close; it need not be brighter on every note. Piezo carries no room return. To compare the captures, lower Room and alternate Close and Piezo, adjusting their levels for similar loudness. **Room Space** selects Wood, Chamber or Hall and is heard when Room is raised. The three capture knobs are independent linear levels: changing one does not rescale the others. Room's fixed voiced response and tail continue running behind its level knob. Lowering all three levels silences the main mix after smoothing; the direct stems remain available.

Drive adds pickup-amplifier saturation to Piezo, including its direct output. It leaves the Close and Room microphone paths unchanged. The main mix passes through stereo **Width**, **Warmth**, a clean output gain, **Limit**, then **Volume** and the final headroom control. Width runs from mono at zero through natural stereo at 1 to wide at 2. Warmth tilts the main mix's low/high balance.

The output stage gives Main L/R a clean **12 dB** gain before Limit, with no change to the stored controls or the instrument's velocity response. The gain follows Drive, so it does not add preamp distortion; limiting reduces the increase on loud hits. Volume keeps its existing response and controls the resulting level.

Turning **Limit** above zero enables the musical limiter; higher values lower its threshold and apply more control. Automatic makeup gain raises the output by as much as the threshold drops, so what the limiter holds down keeps its level while quieter playing and the ringing tails come up: more Limit sounds louder and denser, never quieter. Rear **Attack** (1-50 ms) and **Release** (20-500 ms) shape its linked stereo gain reduction. At zero this stage and its control envelope bypass after a short transition. Small Limit settings no longer round individual peaks through an instantaneous waveshaper.

Separate, always-active headroom control anticipates overloads by **2 ms** and changes gain smoothly to keep all seven outputs within **0.94 sample peak**. Main L/R, Close L/R, Room L/R and mono Piezo each have their own protection. Stereo pairs share gain to preserve their balance; an overloaded capture does not turn down the other outputs. Protection remains active at Limit zero. Once overload recovery has settled, signals below its detector ceiling pass unchanged apart from the delay. All seven outputs receive the same delay, reported to Reason as ``ceil(sample rate × 0.002)`` samples, independently of Quality. The detector also estimates peaks between digital samples to reduce reconstruction overload. The 0.94 bound is a sample-peak limit; it is not a universal intersample/true-peak mastering guarantee. Neither stage removes saturation already produced by Piezo Drive or a physical buzz layer.

Rear connections and setup

Connection or control	Use
Main L/R	Complete processed mix. Connect both for stereo. A single connected main jack carries the mono average.
Close L/R	Close-microphone stems before the Close level knob.
Room (Far) L/R	Far microphones and the voiced room return, both before the Room level knob.
Piezo	Mono contact-pickup stem after Drive, before the Piezo level knob.
Note / Gate CV	Play the instrument from CV. Gate amplitude supplies velocity.
Hardness / Position / Coupling / Sympathetic / Volume CV	Add modulation to the corresponding knob, within its range.
Tune	Global instrument offset, ± 100 cents, added to Reason's master tuning.
Detune	Fixed per-key pitch offsets, up to ± 100 cents at maximum. This differs from the contact variation between hits.
Scale	Equal temperament or the six traditional/measured keyboard choices described below.
Voices	8, 16 or 24 simultaneous voices. Voice stealing fades old voices briefly.
Quality	2x, 4x or 8x internal oversampling. 2x is the factory default. Higher settings use more CPU; 8x at high host sample rates is especially demanding. Changing Quality briefly fades and restarts the engine.

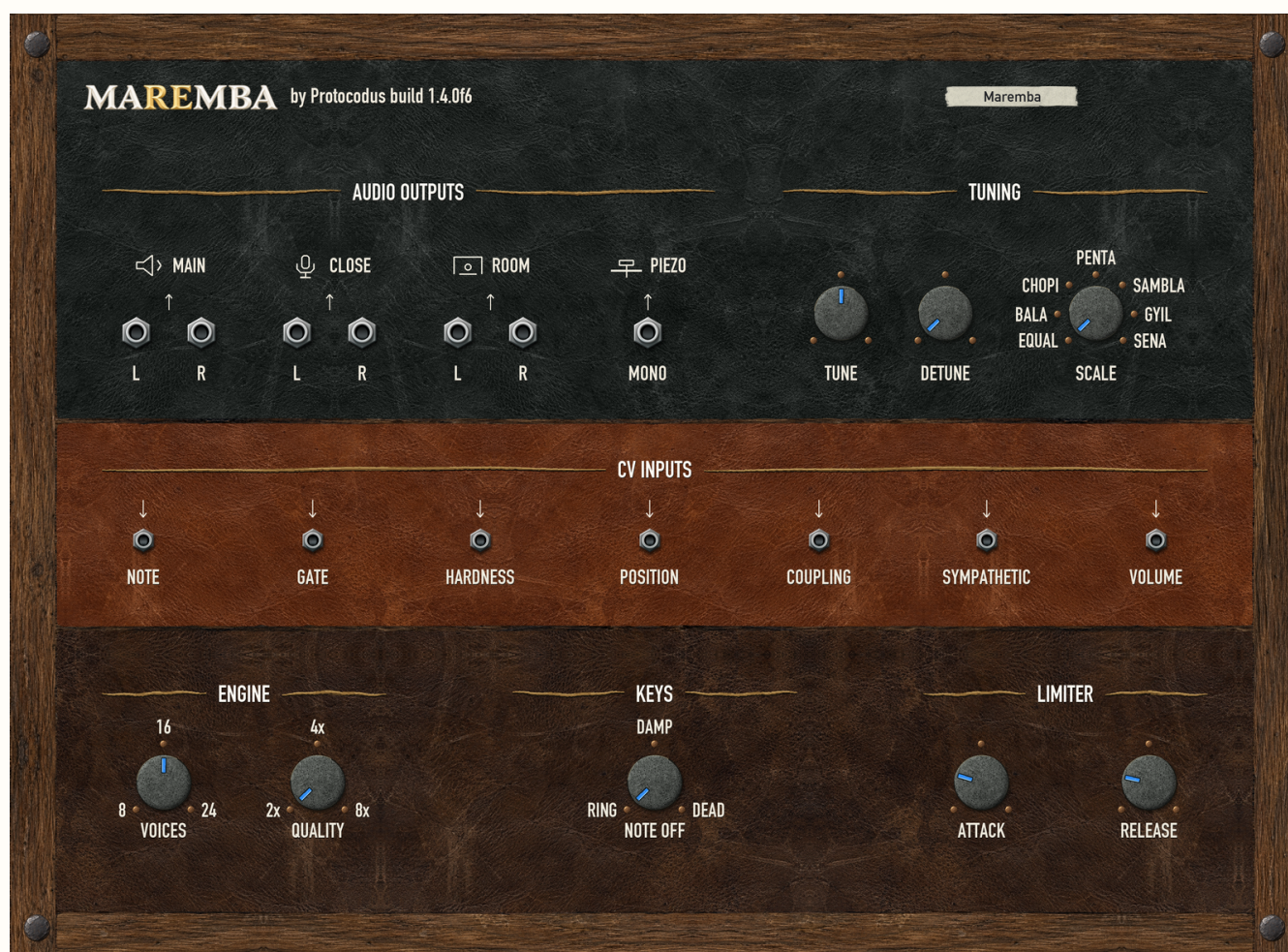
All five direct stems carry a fixed calibration and nothing else: Volume, the Close, Room and Piezo levels, Width, Warmth, the main-output gain and the musical Limit stage shape Main L/R only. Piezo still includes its own Drive. The calibration is set per capture, and for the Kalimba, which is quieter, separately, so that a full-velocity chord reaches the top of every stem alike and its outputs sit level with the bars'; ordinary playing sits well below it. Each capture has its own smooth overload protection, and the matching 2 ms delay keeps them aligned with Main L/R. Changing a capture's front-panel level or the master Volume changes the main mix, while the direct outputs remain independent of them.

Scale's panel abbreviations are **Equal**, **Bala**, **Chopi**, **Penta**, **Sambla**, **Gyl** and **Sena**. They select equal temperament, Bala 7-equal, Chopi timbila, 5-equal pentatonic, Sambla, measured Birifor gyl and measured Sena valimba tunings. Their principal keys follow the respective keyboard layout; intervening MIDI keys are interpolated. New strikes take a changed scale, and sympathetic keys follow it immediately.

Working with existing patches

The factory bank includes 20 patches and the native preset loader remains functional. Stored parameter identities and automation are retained. The older beta's Bow choice plays as Shaft; its replacement factory patch is **Rosewood Roll Chorale**. The hidden legacy Resonator Tune parameter still affects resonator tuning; it is separate from the rear Tune control. The old Glide parameter is retained for compatibility but does not affect sound.

The 1.4 physical-model changes intentionally alter some older patch sounds. Keep an earlier bounce when preserving an exact historical performance matters. A release bundle's evidence report identifies the build, tests and host checks completed for that particular version.



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